

Angles and constructions

— intersecting lines and quadrilaterals

A Cambridge insert: drawing accurately with compasses, ruler and protractor.

LESSON 111–113

3 × 40 MIN

MATEMATIKA 6, QO‘SHIMCHA MAVZU

S7 5.2–5.3

LESSON CLOCK

25'

30'

35'

25'

5

Angles where lines cross	25 min
Constructing triangles	30 min
Constructing quadrilaterals	35 min
Checking	25 min
Homework	5 min

LEARNING OBJECTIVES

- Use the angle facts at intersecting lines.
- Construct a triangle from given sides and angles.
- Construct a rectangle, a parallelogram and a rhombus.
- Check a construction by measuring what was not given.

TEXTBOOK

National	pp. 319–328
Cambridge	Stage 7, pp. 58–65
Workbook	Exercise 5.3

01

Explanation

Angles where lines cross

FACT	STATEMENT
vertically opposite	the two opposite angles are equal
on a line	neighbours add to 180°
at the crossing	all four add to 360°
perpendicular lines	all four are 90°

ONE ANGLE	THE OTHER THREE
70°	$110^\circ, 70^\circ, 110^\circ$
125°	$55^\circ, 125^\circ, 55^\circ$
90°	$90^\circ, 90^\circ, 90^\circ$

TWO CROSSING LINES GIVE ONLY TWO DIFFERENT SIZES

An angle and its supplement, each appearing twice. Knowing one of the four gives all of them.

Constructing triangles

GIVEN	TOOLS	STEPS
three sides	ruler and compasses	draw one side, two arcs, join
two sides and the angle between	ruler and protractor	draw the angle, mark both lengths
a side and two angles	ruler and protractor	draw the side, both angles at its ends

DRAW THE GIVEN ANGLE BEFORE MEASURING THE SIDES

Marking two lengths first and hoping the angle comes out right does not work. The given data fix the order of the construction.

Constructing quadrilaterals

SHAPE	METHOD
rectangle 6×4	two perpendiculars at the ends of a 6 cm side
square of side 5	the same, with equal sides
parallelogram, sides 6 and 4, angle 60°	draw the angle, mark both sides, complete with parallels
rhombus of side 5, angle 70°	as a parallelogram with all sides equal

OPPOSITE SIDES ARE MADE EQUAL, NOT MEASURED TWICE

In a parallelogram, set the compasses to the side length and step it off; measuring each side separately introduces small errors that leave the figure open.

Checking

CONSTRUCTION	CHECK BY MEASURING	SHOULD BE
triangle from three sides	the three angles	they add to 180°
rectangle	the two diagonals	equal
parallelogram	opposite sides	equal
rhombus	the diagonals	they cross at 90°

CHECK SOMETHING YOU WERE NOT GIVEN

Measuring what you drew tells you nothing; measuring a diagonal, or an angle you did not set, is a real test of the construction.

02

Worked examples

EXAMPLE 1 Two lines cross and one angle is 125° . Find the other three.

1

Vertically opposite: 125° .

2

On the line: $180 - 125 = 55^\circ$.

3

Its opposite: 55° .

4

So $125^\circ, 55^\circ, 125^\circ, 55^\circ$.

ANSWER

$55^\circ, 125^\circ, 55^\circ$

EXAMPLE 2 Construct a triangle with sides 6, 5 and 4 cm, and check it.

- ① Draw the 6 cm side.
- ② Arc of 5 from one end, arc of 4 from the other.
- ③ Join the crossing to both ends.
- ④ Measure the three angles: they should total 180° .
The check.

ANSWER

Constructed and checked

EXAMPLE 3 Construct a parallelogram with sides 6 and 4 cm and an angle of 60° .

- ① Draw a 6 cm side.
- ② At one end, set off 60° and mark 4 cm.
- ③ From the two free ends, step off 6 and 4 cm with compasses.
- ④ Check: opposite sides equal, and the other angle is 120° .

ANSWER

Constructed

03 Interactive model

Ask every pupil to measure a diagonal of their rectangle and compare with a neighbour; the two answers agreeing is a better verdict than any teacher's tick.

Crossing lines and constructions

Two lines cross; one angle is 70° . Its opposite is:

Its neighbour is:

How many different sizes appear?

To construct a triangle from three sides you need:

a protractor

compasses

a set square

nothing

To construct one from two sides and the angle between:

compasses only

a protractor and a ruler

a set square

a calculator

A rectangle is checked by measuring:

the sides

the diagonals

the angles only

nothing

A rhombus is checked by:

equal sides

diagonals crossing at 90°

equal diagonals

right angles

A parallelogram with one angle 60° has another of:

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Construction	Yasash	Построение
Intersecting lines	Kesishuvchi chiziqlar	Пересекающиеся прямые
Vertically opposite	Vertikal burchaklar	Вертикальные углы
Parallelogram	Parallelogramm	Параллелограмм
Rhombus	Romb	Ромб
Perpendicular	Perpendikulyar	Перпендикуляр
To check by measuring	O'lchab tekshirish	Проверить измерением
Accuracy	Aniqlik	Точность

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Vertically opposite angles are:

supplementary

equal

complementary

right

The four angles at a crossing add to:

90°

180°

270°

360°

One angle of 125° gives the others:

55, 125, 55

125, 125, 125

35, 125, 35

55, 55, 55

Three given sides need:

a protractor

compasses

a set square

a calculator

A construction should be checked by measuring:

what was given

something not given

nothing

the paper

In a rhombus the diagonals:

are equal

cross at 90°

are parallel

bisect the sides

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. One angle at a crossing is 70° : its opposite

ANSWER 70°

2. And its neighbour

ANSWER 110°

3. One angle is 125° : the neighbour

ANSWER 55°

4. Perpendicular lines: all four angles

ANSWER 90°

5. Angles at a crossing add to

ANSWER 360°

6. Sizes appearing at a crossing

ANSWER 2

7. A parallelogram with one angle 60° : the next

ANSWER 120°

Medium · the standard question

7 PROBLEMS

1. Which tool for three given sides?

ANSWER Compasses

2. Which for two sides and the included angle?

ANSWER Protractor and ruler

3. Which for a side and two angles?

ANSWER Protractor and ruler

4. A rectangle is checked by

ANSWER measuring the diagonals

5. A rhombus is checked by

ANSWER the diagonals crossing at 90°

6. A parallelogram is checked by

ANSWER opposite sides being equal

7. A triangle from three sides is checked by

ANSWER the angles adding to 180°

Hard · stretch and reason

7 PROBLEMS

1. Can a triangle be constructed with sides 3, 4, 8?

ANSWER No — the arcs do not meet

2. A parallelogram of sides 6 and 4 with angle 60° : all four angles

ANSWER $60^\circ, 120^\circ, 60^\circ, 120^\circ$

3. Its perimeter

ANSWER 20 cm

4. A rhombus of side 5 with angle 70° : the other angles

ANSWER $110^\circ, 70^\circ, 110^\circ$

5. Its perimeter

ANSWER 20 cm

6. Why measure a diagonal to check a rectangle?

ANSWER It was not given, so it is an independent test

7. A square of side 5: the diagonal, to the nearest mm

ANSWER 7.1 cm

07 Homework

Homework — 5 tasks

Leave every construction line and arc visible on the page.

1. Two lines cross and one angle is 142° . Find the other three.
2. Construct a triangle with sides 7, 6 and 5 cm and measure its angles.
3. Construct a rectangle 7 cm by 4 cm and measure both diagonals.
4. Construct a parallelogram with sides 7 and 5 cm and an angle of 50° .
5. Construct a rhombus of side 6 cm with one angle 60° and measure its diagonals.